

EE3621: Digital Signal Processing

Lecture 29: DSP for Radar — Pulse Compression, Doppler Processing & CFAR (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Radar Fundamentals: Range, Doppler, and Resolutions.
 - **05:00 – 12:00 (7 mins):** Matched Filter & LFM (Chirp): Waveform equations and Pulse Compression.
 - **12:00 – 18:00 (6 mins):** Pulse Compression Implementation: Stretch processing & FFT.
 - **18:00 – 25:00 (7 mins):** Range-Slow Time Matrix & Doppler Processing: MTI & Pulse-Doppler.
 - **25:00 – 30:00 (5 mins):** Constant False Alarm Rate (CFAR) Detection.
 - **30:00 – 35:00 (5 mins):** FMCW Radar & SAR Basics.
 - **35:00 – 40:00 (5 mins):** Checkpoints & Review.
-

2 Radar Fundamentals

2.1 Range Equation

Radar transmits an electromagnetic pulse and waits for the echo. Let the time-of-flight of the pulse be τ . The round-trip distance is $c\tau$, so the target range R is:

$$R = \frac{c\tau}{2} \quad (1)$$

2.2 Doppler Frequency

If a target is moving with radial velocity v , the round trip distance changes as $2vt$, introducing a phase shift $\phi(t) = -\frac{2\pi(2vt)}{\lambda}$. The Doppler frequency shift is:

$$f_d = -\frac{1}{2\pi} \frac{d\phi}{dt} = \frac{2v}{\lambda} \quad (2)$$

2.3 Resolutions

The range resolution δR and Doppler resolution δv are:

$$\delta R = \frac{c}{2B} \quad (3)$$

$$\delta v = \frac{\lambda}{2T_{\text{dwell}}} \quad (4)$$

3 Matched Filter for Radar

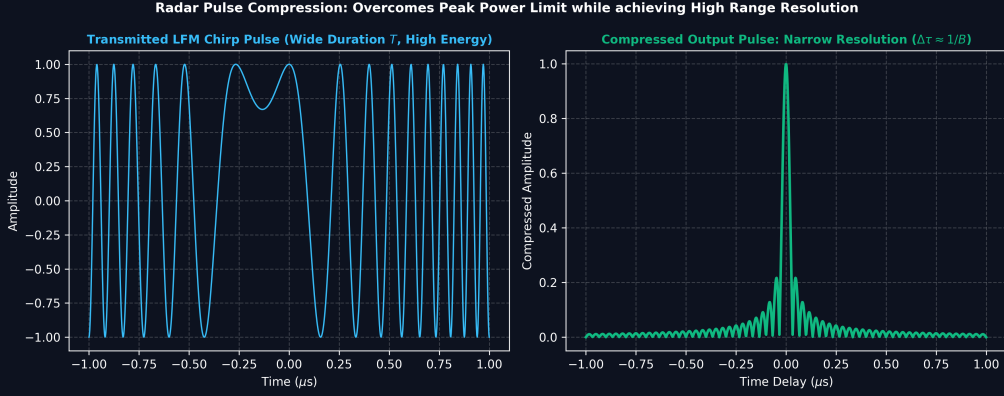


Figure 1: Radar Linear Frequency Modulation (Chirp) Pulse Compression for High Range Resolution.

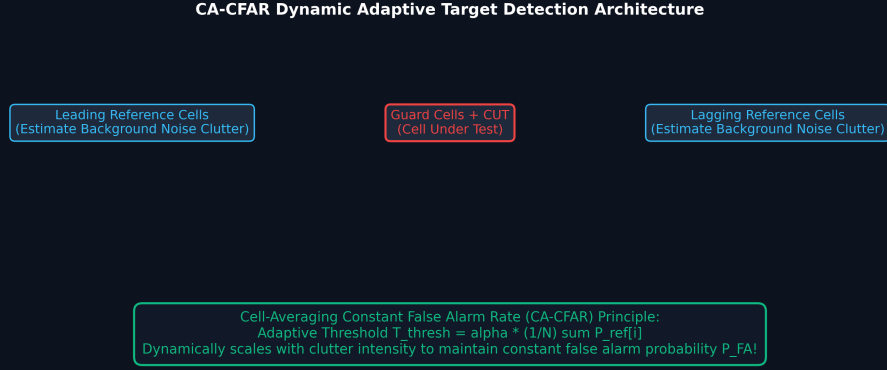


Figure 2: Cell-Averaging Constant False Alarm Rate (CA-CFAR) Adaptive Threshold Target Detection.

3.1 Linear Frequency Modulated (LFM) Chirp

The LFM chirp signal is defined as:

$$s(t) = \text{rect}(t/T) e^{j\pi\mu t^2} \quad (5)$$

where $\mu = B/T$ is the chirp rate.

3.2 Matched Filter Response

The matched filter impulse response is:

$$h(t) = s^*(-t) \quad (6)$$

The output pulse compression is:

$$|R_{ss}(\tau)|^2 = T^2 \text{sinc}^2(\pi B\tau) \quad (7)$$

The Time-Bandwidth Product (TBP) represents the compression ratio:

$$\text{TBP} = B \cdot T \quad (8)$$

4 Pulse Compression Implementation

We use the Frequency Domain for fast processing:

$$Y(f) = R(f)H^*(f) = R(f)S^*(f) \quad (9)$$

This is implemented as an FFT-Multiply-IFFT structure. For extreme bandwidths, stretch processing mixes the signal with a reference chirp.

5 Range Cell Processing

A radar transmits N_{chirps} pulses. We stack each received pulse into a 2D **Range-Slow Time Matrix**, where rows are range bins and columns are successive pulses.

6 Doppler Processing (MTI & Pulse-Doppler)

6.1 Moving Target Indicator (MTI)

MTI subtracts consecutive pulses to cancel stationary clutter:

$$y(n) = x(n) - x(n-1) \quad (10)$$

In designing more complex MTI filters, analog prototype filters can be digitized using the Matched Z-Transform (MZT) (Figure ??).

Spectral transformations can adapt the MTI notch (Figure ??).

6.2 Pulse-Doppler Processing

A 2D FFT over the Range-Slow Time Matrix produces a **Range-Doppler Map**, providing both distance and velocity.

7 Constant False Alarm Rate (CFAR) Detection

The threshold is $T_{thresh} = T \cdot \bar{P}_{noise}$. For Cell Averaging CFAR (CA-CFAR) with N reference cells and false alarm probability P_{fa} , the multiplier is:

$$T = N \left(P_{fa}^{-1/N} - 1 \right) \quad (11)$$

8 FMCW Radar & SAR Basics

8.1 FMCW Radar

The beat frequency for a stationary target is:

$$f_b = \frac{2R}{c} \frac{B}{T} \quad (12)$$

8.2 SAR Basics

Synthetic Aperture Radar produces high-resolution images. The azimuth resolution is surprisingly independent of range:

$$\delta a = \frac{D}{2} \quad (13)$$

9 Checkpoint & Quick Review Questions

Q1: Calculate range resolution and TBP for $B = 50$ MHz and $T = 20\mu s$.

Answer: $\delta R = \frac{c}{2B} = 3$ m. $TBP = 50 \times 10^6 \times 20 \times 10^{-6} = 1000$.

Q2: Find CFAR threshold T for $N = 16$ and $P_{fa} = 10^{-6}$.

Answer: $T = 16((10^{-6})^{-1/16} - 1) = 16(10^{0.375} - 1) \approx 21.94$.

Q3: An FMCW radar with $B = 1$ GHz and $T = 100\mu s$ has a beat frequency of 150 kHz. Find range R .

Answer: $R = \frac{f_b c T}{2B} = \frac{150 \times 10^3 \times 3 \times 10^8 \times 100 \times 10^{-6}}{2 \times 10^9} = 2.25$ m.